

Estimates of relative contribution of biomass, harvest index and yield components to soybean yield improvements in China

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Abstract

A method was developed to estimate the relative contribution of increased biomass, harvest index and yield components to seed yield gain of soybean. Data from four field experiments conducted in three soybean production regions in China were used to measure the role of different traits (such as harvest index) in seed yield improvement by applying this method. The results indicated that harvest index was a larger contributor to the progress of soybean yield improvements than biomass in China. However, the estimates of improvements of yield components suggested that there could be several strategies for obtaining high yield. High yielding would be achieved by improving all of the three yield components (pods per plant, seeds per pod and seed weight) simultaneously, or significantly improving one or two of the three components.

Key words: biomass — *Glycine max* — harvest index — yield components — yield improvement

Soybean [*Glycine max* (L.) Merr.] is an important crop in many diverse regions of the world, especially in China where it has been cultivated for more than 5000 years. Yield gains of soybean were 12.1 kg/ha/year from 1950 to 1991 in China (Hu 1994) and 22.6 kg/ha/year from 1924 to 1997 in the USA (Specht et al. 1999). In the USA, linear estimates of the annual genetic improvements in soybean yield were in the range of about 10–30 kg/ha/year (Specht et al. 1999). Ye et al. (1996) tested 23 cultivars grown in Huanghuaihai region (a main soybean production region in China) released from the early 1950s to the late 1980s and it was found that the genetic gain of seed yield was estimated to be from 1.2 to 2.5% annually. Voldeng et al. (1997) evaluated 41 cultivars released in Ontario, Canada over a 58-year period and their linear estimate of the annual genetic improvements was 11.1 kg/ha/year, although they noted that yield improvement prior to 1976 was essentially zero and 30 kg/ha/year thereafter. Karmakar and Bhatnager (1996) reported the genetic gain in seed yield of soybean varieties in India during 1969–1993 to be approximately 22 kg/ha (1.2% per annum).

Experiments using historical soybean cultivars grown in the same environments may offer researchers an opportunity to measure yield progress and to determine its cause. Correlation and regression analysis were usually used to determine the relationship between seed yield and biomass, harvest index and yield components (Ye et al. 1996, Morrison et al. 1999, 2000). However, there has not been a method to measure the relative contribution of biological yield, harvest index and yield components to soybean seed yield. In this paper just such a method is described.

Materials and Methods

Experimental data: To measure the relative contribution of biological yield, harvest index and yield components to yield improvement, data from four experiments were used. These data included improvements of three growth habit types in Liaoning Province (experiment 1; Tian et al. 1988), improvements in the Huaibei region of Jiangsu Province (experiment 2; Feng et al. 1991), improvements in the Huanghuaihai region (experiment 3; Ye et al. 1996) and improvements from hybridization in Liaoning Province (experiment 4; Pei et al. 1997). More details are listed in Table 1.

Measures were conducted on 6, 16, 23 and 19 soybean cultivars in experiments 1 to 4, respectively. The experiments were planted in three- to eight-row plots in a randomized block design with three or four replications. The rows were 3–5 m long and 0.4–0.6 m apart. A sample of 10 plants was harvested from each plot at final harvest to record pods and seeds and calculate pods per plant and seeds per pod. Seed weight was measured on two 100-seed random samples. The biomass, seed yield and harvest index were calculated from the middle row of each plot (biomass and harvest index in experiment 3 were not measured). The above-ground plant material was dried to a constant weight and its weight was recorded before thrashing (fallen leaves and petioles were included in the biomass in experiment 2). After thrashing, the seed yield was recorded and used for yield calculation. The harvest index was calculated as seed yield/weight of dried above-ground plant material (Tian et al. 1988, Feng et al. 1991, Ye et al. 1996, Pei et al. 1997).

Relative contribution of biomass and harvest index: The basic equation is $Y_e = HI \times Y_b$, where Y_e , HI and Y_b indicate economic yield, harvest index and biomass, respectively. Then at timepoint t_0 $Y_{e0} = HI_0 \times Y_{b0}$ and at t_1 a change to $Y_{e1} = HI_1 \times Y_{b1}$. This yields

$$\frac{Y_{e1}}{Y_{e0}} = \frac{HI_1 \times Y_{b1}}{HI_0 \times Y_{b0}}$$

Using natural logarithms on both sides yields

$$\ln \frac{Y_{e1}}{Y_{e0}} = \ln \frac{HI_1}{HI_0} + \ln \frac{Y_{b1}}{Y_{b0}}$$

After division by $\ln \frac{Y_{e1}}{Y_{e0}}$

$$\frac{\ln \frac{HI_1}{HI_0}}{\ln \frac{Y_{e1}}{Y_{e0}}} + \frac{\ln \frac{Y_{b1}}{Y_{b0}}}{\ln \frac{Y_{e1}}{Y_{e0}}} = C_{HI} + C_{Y_b} = 1$$

where C_{HI} and C_{Y_b} are the contributions of increased harvest index and biological yield to higher yield.

Relative contribution of yield components: According to the equation $Y_e = KN_p N_g W_g$, where K , N_p , N_g and W_g are the correction coefficient

Table 1: Description of experiments from which data were used in this study

Experiment	Representative soybean production region	Period and cultivars	Source
Experiment 1	Liaoning Province of Northeast region	Semi-determinate: Changnong 2 (released in early 1980s) Xiaojinhuang 1 (local variety in 1950s) Indeterminate: Jilin 3 (released in early 1980s) Tiejiasilihuang (local variety in 1950s) Determinate: Gongjiao 7514-2 (released in early 1980s) Zaoshufengdihuang (local variety in 1950s)	Tian et al. 1988
Experiment 2	Huaibei region of Jiangsu Province	1980s: four varieties (Youbian 30, Huaidou 1, Guandou 1 and Sidou 11) also derived from cross-breeding 1970s: four varieties (Xudou 1, Xudou 2, Xudou 3 and Xuzhou 135) derived from cross and pedigree selection 1960s: four varieties (Xuzhou 301, Xuzhou 302, 58-161 and Guanyuan 1) derived from cross-breeding 1950s: four local varieties (Yancheng Xiaoyoudou, Guannan Baihuazi, Guanyun Maoyedou and Xuyang Pingdinghuang)	Feng et al. 1991
Experiment 3	Huanghuaihai region including Shandong, Hebei, Henan, Shanxi Province and Beijing	1980s: seven varieties (Yudou 8, Zhonghuang 2, Kefeng 6, Jindou 7, Jindou 9, Zhonghuang 3 and Jindou 17) derived from cross-breeding 1970s: six varieties (Zhengzhou 135, Yuejin 5, Youbian 30, Jindou 4 Ludou 4 and Jidou4) derived from cross-breeding 1960s: five varieties (Shanjindou, Tongxian yuandou, Xuan 23, Zaofeng 1 and Wenfeng 5) selected from previous local varieties by pedigree method 1950s: five local varieties (Niumaohuang, Tiejiaohuan, Pingdinghuang, Taiguzao and Guangping Niumaohuan)	Ye et al. 1996
Experiment 4	Liaoning Province of Northeast region	All experimental varieties were derived from cross-breeding 1985-1993: eight varieties (Kaiyu 9, and 10, Tiefeng 21, 22, 24, and 26, Liaodou 9 and 10) 1975-1984: six varieties (Dandou 3, 4, and 5, Kaiyu 8, Tiefeng 20, and Liaodou 3) 1967-1974: five varieties (Tiefeng 8, 9, 18, and 19, and Dandou 2)	Pei et al. 1997

the number of pods per plant, seed number per pod and seed weight respectively. The relative contribution of the three yield components to yield gain could be estimated as shown above, yielding

$$C_{N_p} = \frac{\ln \frac{N_{p1}}{N_{p0}}}{\ln \frac{Y_{e1}}{Y_{e0}}}, C_{N_g} = \frac{\ln \frac{N_{g1}}{N_{g0}}}{\ln \frac{Y_{e1}}{Y_{e0}}}, C_{W_g} = \frac{\ln \frac{W_{g1}}{W_{g0}}}{\ln \frac{Y_{e1}}{Y_{e0}}}$$

When K values at the two breeding periods are the same, then $C_{N_p} + C_{N_g} + C_{W_g} = 1$. Otherwise, a corrected weight C' is obtained by a weighting factor, $a = 1/(C_{N_p} + C_{N_g} + C_{W_g})$.

Results

Relative contribution of biomass and harvest index to yield gain

Yield, biomass, harvest index and the relative contribution to yield gain are listed in Table 2. Seed yield was improved with both biomass and harvest index over time in experiments 1, 2 and 4. For example, yield, biomass and harvest index gains were 144.9 g/m², 67.2 g/m² and 0.184 respectively in experiment 2 during the early 1950s to the late 1980s. This indicated that both biomass and harvest index had positive effects on soybean yield improvement. However, the relative contribution of harvest index to yield gain was larger than that of biomass in the progress of yield improvement (Table 2). In experiment 1, the average of the relative contribution (novel cultivars to previous cultivars) of harvest index of three growth types was 67.4%, twice of that of biomass. In experiments 2 and 4, as soybean yield improved progressively there were the same decrease trends in the relative contributions of harvest

index to yield, whereas the relative contributions of biomass to yield increased moderately.

Relative contribution of yield components to yield gain

Results of relative contribution of yield components to yield gain are summarized in Table 3. In experiment 1, semi-determinate and determinate varieties showed similar trends for improvement, seed weight being the main contributor to yield improvement, and there was little or negative contribution from pods per plant and seeds per pod. For indeterminate types the relative contribution of pods per plant was larger than that of both seeds per plant and seed weight which had similar contributions. In experiments 2 and 3, seed weight was the largest contributor, but in experiment 4 pods per plant was the largest contributor to yield gain. There were two types of yield component improvements; all yield components had positive but different contributions to yield gain in experiments 2 and 4, but in experiment 3 there were two yield components (seeds per pod and seed weight) which had positive contributions and one yield component (pods per plant) which had a negative contribution to yield gain.

Discussion

Basically there are two major ways to achieve higher economic yields: by increasing harvest index and/or by increasing biomass (biological yield). Biomass is the basis for obtaining higher seed yield. Genetic improvements of grain yield in

Table 2: Measures of relative contribution of biological yield and harvest index to yield gain

Experiment	Periods varieties released	Yield (g/plant)	Biomass (g/plant)		Harvest index	
			Mean	$C_{Y_b}/\%$	Mean	$C_{HI}/\%$
Experiment 1	1980s	24.3	56.5	36.0	0.430	64.0
		22.1			0.405	
	1950s	21.0	50.5	25.0	0.416	75.0
		16.3			0.344	
	1980s	25.7	61.6	36.7	0.417	63.3
		21.4			0.372	
	1980s	15.2	37.5	17.8	0.405	82.2
		13.1			0.366	87.8
Experiment 2	1970s	12.8	36.1	15.3	0.355	84.6
	1960s	9.0	34.2		0.263	
Experiment 4	1985–1993	21.7	37.9	45.6	0.574	54.4
	1975–1984	21.0	35.0	10.2	0.600	89.8
	1967–1974	17.6	34.4		0.512	

Table 3: Measures of relative contribution of yield components to yield gain

Experiment	Periods varieties released	Yield (g/plant) (see Table 2)	Number of pods per plant		Number of seeds per pod		Seed weight (g)	
			Mean	$C_{N_p}/\%$	Mean	$C_{N_s}/\%$	Mean	$C_{W_s}/\%$
Experiment 1	1980s	24.3	74.5	-33.9	2.31	7.7	18.1	126.2
		22.1	78.2		2.28		15.1	
	1950s	21.0	63.4	45.0	2.56	29.2	16.3	25.8
		16.3	56.7		2.40		15.2	
	1980s	25.7	67.9	2.2	1.91	-3.6	24.8	101.4
		21.4	67.7		1.92		21.6	
	1980s	15.2	42.7	14.5	2.03	14.0	18.0	71.5
		13.1	40.4		2.12	26.7	15.9	66.1
Experiment 2	1970s	12.8	43.2	19.2	1.97	10.8	16.7	70.0
	1960s	9.0	39.0		1.86		11.5	
Experiment 3	1980s	20.3	49.4	-101.3	2.30	66.7	19.4	134.6
	1970s	18.6	53.6	-90.2	2.25	75.8	17.2	114.4
	1960s	16.9	64.1	5.7	2.05	30.3	15.4	64.0
	1950s	14.6	63.5		1.95		13.9	
Experiment 4	1985–1993	21.7	77.7	52.8	1.70	14.2	18.6	33.0
	1975–1984	21.0	74.9	69.0	1.66	8.6	17.7	22.4
	1967–1974	17.6	68.0		1.64		17.2	

wheat, rice, barley, peanut and soybean have been closely associated with increases in harvest index (Gifford et al. 1984). On the other hand, yield component selections have played direct roles on yield improvements.

This study indicated that harvest index was a larger contributor to the progress of soybean yield improvements in China compared with biomass. This is consistent with the results obtained in Canada and India (Karmakar and Bhatnager 1996, Morrison et al. 1999). Karmakar and Bhatnager (1996) reported that the varieties of selection cycle 2 showed a 19% higher seed yield and a 16% increase in harvest index over selection cycle 1. Morrison et al. (1999) found that there was no significant change in total dry weight (biomass) produced by old and new cultivars, and harvest index increased linearly by 0.47% per year with the year of release. Physiological evaluation of old vs. new soybean cultivars indicated that yield improvement was associated with

assimilate supply during the seed-filling period (Shiraiwa and Hashikawa 1995, Kumudini et al. 2001). From Table 2 there is a decreased trend in the relative contribution of harvest index, a slight effect in experiment 2 and a significant one in experiment 4. Assuming that the harvest index is already high for most modern cultivars, it appears that there is not much potential for further improvement of this trait (Sinclair 1997). Ma et al. (1995) suggested that seed yield had a significantly positive correlation with biological yield, according to experimental data from 1973 to 1982, and in 1993.

Relative contribution estimates of yield components to soybean yield gain showed that there could be different strategies to obtain high yield. Higher yields could be obtained by improving all three yield components, or just one or two of the three components. Boerma (1979) reported no significant correlation of yield improvement across time with any measured yield component. Morrison et al. (2000) indicated that

plant breeders in short-season regions have increased soybean yield by increasing the number of seeds per plant, not by increasing seed size. It did not appear that the increase in seeds per plant occurred at the expense of seed size. Egli et al. (1978) found that yield was correlated with both seed number per plant and seed weight. They stated that seed weight was phenotypically more unstable than seed number per plant. Sui et al. (1986) reported significant correlation of yield improvement across time with seed size in the Helongjiang Province of China. In India, seed yield showed a significant positive association with mean seed weight ($r = 0.65$) (Karmakar and Bhatnager 1996).

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